

Cosmological Vacuum Selection and the Maximal Chronometric-Shear Ridge

Reheating, replicated-sector asymmetry, the QCD crossover, matter locking, and parameter optimization in Z6 QCD chronometry

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1 Executive result

The cosmological calculation changes the preferred parameter region, but does not kill the model.

The old local-screening benchmark,

$$N = 6, \quad M = 10 \text{ TeV}, \quad f_a = M_{\text{Pl}}, \quad \epsilon = 10^{-6},$$

has negligible thermal response:

$$\eta_{\Psi, \text{max}} = 2.27 \times 10^{-7}, \quad \eta_{\text{QCD}, \text{max}} = 2.81 \times 10^{-7}.$$

The early Universe therefore cannot select its vacuum. With a generic displacement of order $\pi/6$, its constant-mass misalignment estimate is

$$\Omega_a h^2 \simeq 1.6 \times 10^2.$$

That benchmark remains useful for local Earth-Sun screening, but it is not a satisfactory generic cosmological benchmark.

A viable cosmological region instead lies on a lower- f_a , smaller- ϵ ridge. The strongest fully evolved benchmark found here is

$$N = 6, \quad f_a = 2.435 \times 10^{10} \text{ GeV}, \quad \epsilon = 2.70 \times 10^{-13}, \quad M = 1.002 \times 10^6 \text{ GeV}$$

with

$$T_R = 100M = 1.002 \times 10^8 \text{ GeV}, \quad m_a = 7.5 \times 10^{-29} \text{ eV}, \quad d_g = 10^{-6}.$$

One adjacent replica is populated at reheating with temperature ratio

$$\xi \equiv T_5/T_0 = 0.25.$$

This gives

$$\Delta N_{\text{eff}} \simeq 0.0289,$$

below the current 95% upper limit $\Delta N_{\text{eff}} < 0.107$. The heavy threshold and QCD crossover both dominate Hubble friction,

$$\eta_{\Psi, \text{max}} = 612, \quad \eta_{\text{QCD}, \text{max}} = 759,$$

and all 12 evenly spaced homogeneous initial phases tested numerically converge to the same adjacent vacuum,

$$x_v = \frac{\pi}{6}, \quad x \equiv a/f_a.$$

The misalignment abundance is negligible,

$$\Omega_a h^2 \simeq 2.3 \times 10^{-18}.$$

The maximum generic chronometric signal is not set by cosmology. It remains limited by equivalence-principle constraints. Taking the conservative working value $d_g = 10^{-6}$, the predicted solar annual peak-to-peak clock modulation is

$$\left| \Delta \ln \frac{\nu_A}{\nu_B} \right|_{\text{yr}} \simeq 6.58 \times 10^{-22} |K_{AB}|.$$

For an ordinary atomic sensitivity $|K_{AB}| \sim 1$, this is remote. For the provisional nuclear sensitivity $|K| \sim 10^4$ used in recent thorium-229 analyses, it becomes 6.6×10^{-18} , although the nuclear matrix-element uncertainty is substantial.

The main conceptual result is:

cosmology can select the sign and branch of chronometric shear, but it does not raise its allowed magnitude above the EP bo

2 Model

2.1 Zero-temperature protected potential

The compact ratio mode is

$$x = \frac{a}{f_a}.$$

The replicated vectorlike thresholds are

$$M_k(x) = M \left[1 - \epsilon \cos \left(x + \frac{2\pi k}{N} \right) \right], \quad k = 0, \dots, N-1.$$

The cyclic orbit cancels all lower harmonics in the vacuum effective potential. For $N > 4$,

$$V_0(x) = \frac{m_a^2 f_a^2}{N^2} (1 + \cos Nx),$$

where

$$m_a^2 = \frac{N_c N^2 F_N}{8\pi^2} \frac{M^4}{f_a^2} \epsilon^N,$$

and

$$F_N = \frac{24 \, 2^{1-N}}{(N-1)(N-2)(N-3)(N-4)}.$$

The minima are

$$x_v = \frac{(2j+1)\pi}{N}.$$

The visible QCD threshold transmits the ratio-mode response as

$$d \ln \frac{\Lambda_3}{\chi} = \left[\frac{2}{27} + O(\alpha_s) \right] d \ln \frac{M_0}{\chi}.$$

At a selected vacuum,

$$d_g = \frac{2}{27} \frac{M_{\text{Pl}}}{f_a} \frac{\epsilon \sin x_v}{1 - \epsilon \cos x_v}.$$

2.2 Cosmological equation

The homogeneous field obeys

$$\ddot{x} + 3H\dot{x} + \frac{1}{f_a^2} \frac{\partial V_{\text{eff}}}{\partial x} = 0,$$

with

$$V_{\text{eff}} = V_0 + V_{\Psi,T} + V_{\text{QCD},T} + V_b.$$

The numerical evolution includes visible and hidden radiation, standard matter, and a cosmological constant in H .

3 Reheating and the heavy-threshold thermal potential

For a Dirac colour triplet with $g_{\Psi} = 12$, the exact one-loop thermal free-energy contribution is

$$V_{\Psi,k}(T_k, x) = -\frac{g_{\Psi} T_k^4}{2\pi^2} \int_0^{\infty} dq q^2 \ln \left[1 + e^{-\sqrt{q^2 + M_k^2(x)/T_k^2}} \right].$$

At leading order in ϵ ,

$$V_{\Psi,T}^{(1)} = -\sum_k A_{\Psi}(T_k) \cos \left(x + \frac{2\pi k}{N} \right),$$

where

$$A_{\Psi}(T) = \epsilon M \frac{\partial V_{\Psi}(T, M)}{\partial M}.$$

For $T \gg M$,

$$A_{\Psi}(T) \longrightarrow \frac{g_{\Psi}}{24} \epsilon M^2 T^2.$$

Each decoupled replica conserves entropy independently:

$$T_k g_{*s}^{1/3}(T_k) = \xi_k T_0 g_{*s}^{1/3}(T_0).$$

At high temperature, define the thermal phasor

$$W_2 = \sum_k \xi_k^2 e^{2\pi i k/N}.$$

Then

$$V_{\Psi,T}^{(1)} \propto -\text{Re} (e^{ix} W_2), \quad x_T = -\arg W_2.$$

For the benchmark population,

$$\xi_0 = 1, \quad \xi_5 = 0.25,$$

and all other replicas are empty. Therefore

$$W_2 = 1 + 0.25^2 e^{-i\pi/3}, \quad \boxed{x_T = 0.0524383}.$$

This is a useful economy: the branch-selecting phase scales as ξ^2 , while the dark-radiation cost scales as ξ^4 . For a complete hidden Standard Model copy with its standard internal neutrino-photon entropy history,

$$\Delta N_{\text{eff}} \simeq 7.403 \xi^4.$$

Thus $\xi = 0.25$ gives

$$\Delta N_{\text{eff}} = 0.0289.$$

Only one adjacent copy can be populated at this level: five copies at $\xi = 0.25$ would give $\Delta N_{\text{eff}} \simeq 0.145$, above the current bound.

4 QCD crossover

For QCD pressure written as

$$p(T, \Lambda) = T^4 P(T/\Lambda),$$

the thermodynamic identity

$$\left. \frac{\partial F}{\partial \ln \Lambda} \right|_T = \rho - 3p \equiv \Theta(T)$$

follows directly from dimensional analysis.

The threshold relation gives

$$\Lambda_k(x) = \Lambda_0 \left[1 - \epsilon \cos \left(x + \frac{2\pi k}{N} \right) \right]^{2/27}.$$

The leading lock-defect contribution is therefore

$$V_{\text{QCD},k}(T_k, x) = \frac{2}{27} \Theta(T_k) \ln \left[1 - \epsilon \cos \left(x + \frac{2\pi k}{N} \right) \right].$$

The evolution uses the 2 + 1 + 1-flavour lattice trace-anomaly fit

$$\frac{\Theta(T)}{T^4} = e^{-h_1/t - h_2/t^2} \left[h_0 + f_0 \frac{\tanh(f_1 t + f_2) + 1}{1 + g_1 t + g_2 t^2} \right], \quad t = \frac{T}{0.2 \text{ GeV}},$$

with

$$(h_0, h_1, h_2, f_0, f_1, f_2, g_1, g_2) = (0.353, -1.04, 0.534, 1.75, 6.80, -5.18, 0.525, 0.160).$$

For the strong benchmark, the QCD focusing measure peaks at

$$T \simeq 0.179 \text{ GeV}, \quad \eta_{\text{QCD}} \equiv \frac{4m_{T,\text{QCD}}^2}{H^2} = 759.$$

The hidden sector crosses its own QCD region at a different visible temperature. This produces a time-dependent phasor rather than a single fixed thermal minimum.

5 Matter domination

Nonrelativistic visible baryons contribute

$$V_b(x, a) = \rho_b(a) (1 - \epsilon \cos x)^{p_b}, \quad p_b \simeq \frac{2}{27}.$$

Near the even- N thermal maximum $x = 0$, this gives a positive curvature

$$m_b^2(a) = \frac{\rho_b(a) p_b \epsilon}{f_a^2}.$$

The zero-temperature potential gives a tachyonic curvature $-m_a^2$. The homogeneous branch is released when

$$m_b^2(a) = m_a^2.$$

Thus

$$1 + z_{\text{inst}} = \left[\frac{m_a^2 f_a^2}{\rho_{b0} p_b \epsilon} \right]^{1/3}.$$

For the moderate and strong benchmarks,

$$z_{\text{inst}} = 1224, \quad z_{\text{inst}} = 450,$$

respectively. Baryons do not screen the scalar in finite bodies, as shown in v0.7, but the homogeneous cosmic baryon background can delay the roll and alter the selected branch.

6 Vacuum selection

6.1 Moderate-focusing regime

The moderate point is

$$f_a = 4.87 \times 10^{11} \text{ GeV}, \quad \epsilon = 5.40 \times 10^{-12}, \quad M = 5.01 \times 10^4 \text{ GeV}.$$

It has

$$\eta_{\Psi, \text{max}} = 30.6, \quad \eta_{\text{QCD}, \text{max}} = 38.0.$$

Nevertheless, a 12-point scan over the inflationary homogeneous phase gives five trajectories in $+\pi/6$ and seven in $-\pi/6$. The sudden disappearance of thermal curvature and subsequent QCD and matter-era phase evolution preserve information about the initial zero mode.

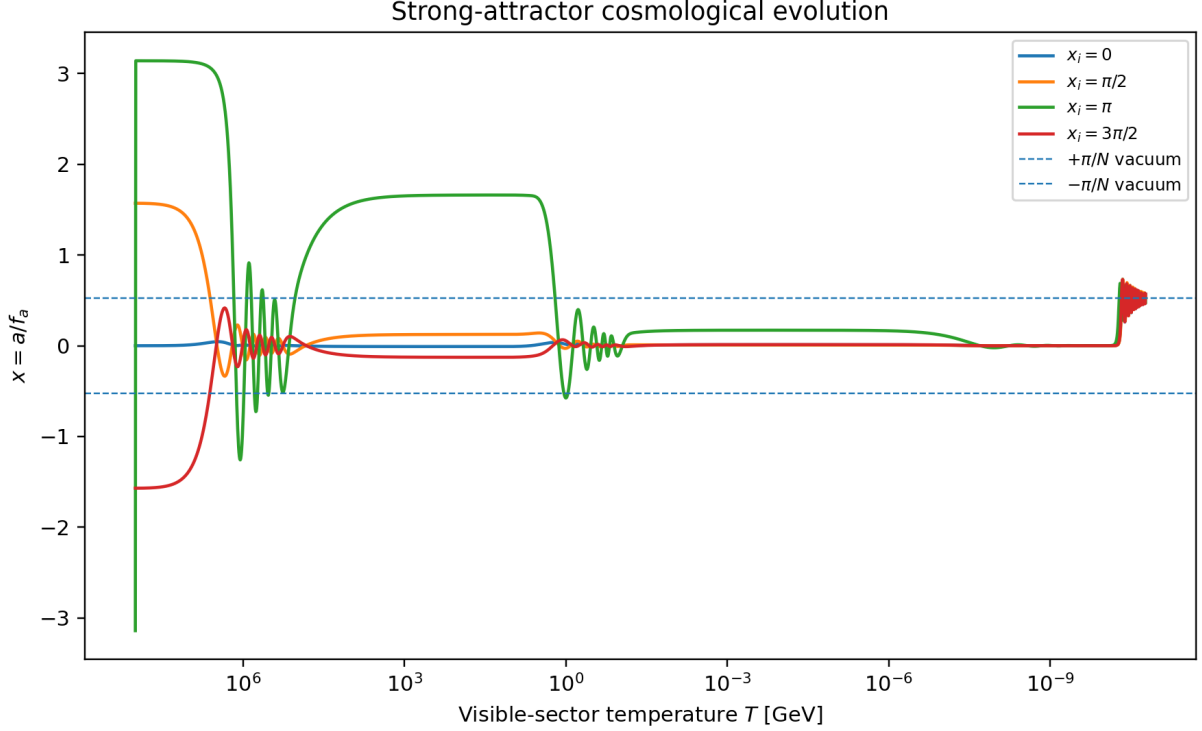


Figure 1: Strong-attractor trajectories. Four widely separated homogeneous initial phases converge to the same adjacent vacuum after heavy-threshold focusing, QCD refocusing, baryon locking, and late release.

6.2 Strong-attractor regime

The strong point is

$$f_a = 2.435 \times 10^{10} \text{ GeV}, \quad \epsilon = 2.70 \times 10^{-13}, \quad M = 1.002 \times 10^6 \text{ GeV}.$$

It has

$$\eta_{\Psi, \max} = 612, \quad \eta_{\text{QCD}, \max} = 759.$$

All 12 evenly spaced homogeneous initial phases tested converge to $+\pi/6$. The largest residual distance from that vacuum at the numerical stopping surface is 0.049 rad.

This is a numerical existence result, not a theorem that every point with smaller f_a is a global attractor. Branch selection is phase-sensitive and can form islands in parameter space.

6.3 Domain-wall condition

The exact zero-temperature theory retains Z_6 -degenerate vacua. The clean cosmology therefore requires:

1. the approximate $U(1)$ to break before or during inflation;
2. no post-inflation restoration, conservatively $T_R < f_a$;
3. inflationary phase fluctuations smaller than the reheating bias.

Since

$$\delta x_{\text{inf}} \simeq \frac{H_{\text{inf}}}{2\pi f_a},$$

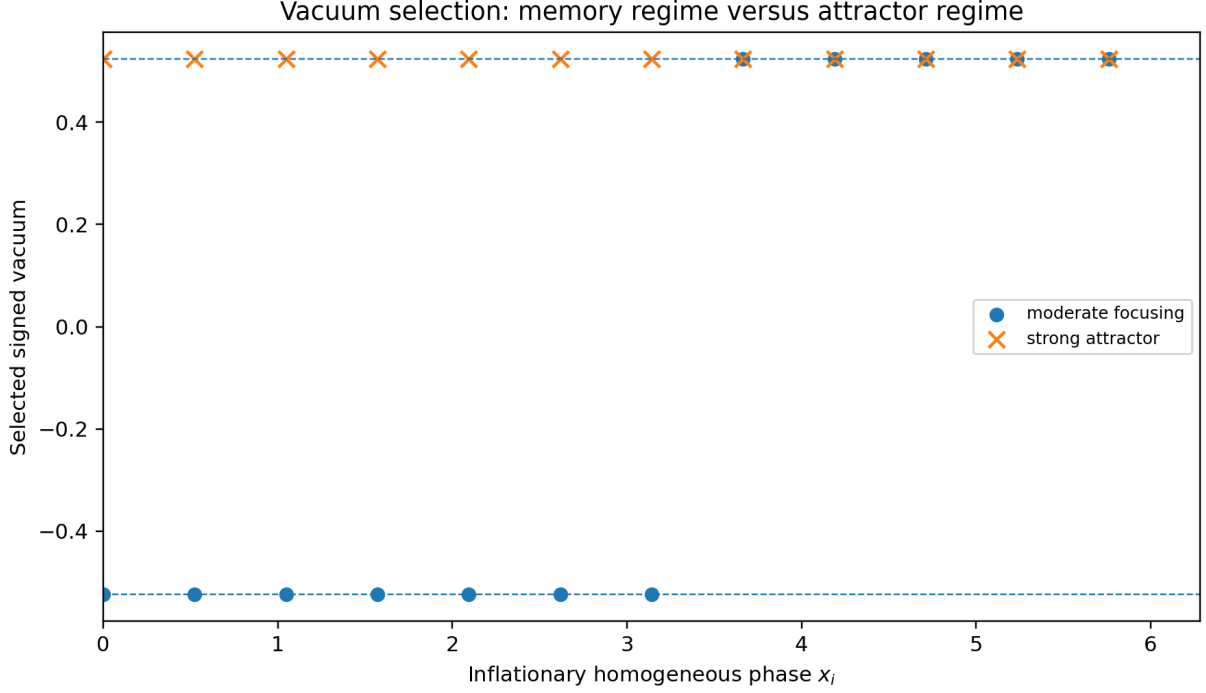


Figure 2: The same reheating asymmetry produces a two-branch memory regime at moderate focusing and a one-branch attractor in the tested strong benchmark.

a useful sufficient condition is

$$H_{\text{inf}} \ll 2\pi f_a x_T.$$

For the strong benchmark,

$$2\pi f_a x_T = 8.0 \times 10^9 \text{ GeV}.$$

If the symmetry is restored after inflation, an additional technically safe permanent bias is required to remove strings and stable domain walls. The transient reheating asymmetry alone is not a permanent wall-lifting term.

7 Why N=6 is preferred

Visible-dominated reheating with even N focuses the field near a zero-temperature maximum. The nearest minima are

$$x_v = \pm \frac{\pi}{N}.$$

For $\epsilon \ll 1$,

$$d_g \simeq \frac{2}{27} \frac{\epsilon}{y} \sin \frac{\pi}{N}, \quad y \equiv \frac{f_a}{M_{\text{Pl}}}.$$

At fixed d_g ,

$$\epsilon = \frac{27}{2} \frac{d_g y}{\sin(\pi/N)}.$$

The smallest even $N > 4$ is $N = 6$, which simultaneously gives:

- the largest adjacent-vacuum slope among allowed even N ;
- the fewest replicated sectors;
- the lowest heavy threshold required for a chosen m_a ;
- the least severe reheating architecture.

For $N = 6$,

$$\boxed{\epsilon = 27 d_g y}.$$

The exact coloured-fermion thermal function gives

$$\eta_{\Psi, \max} \simeq 0.22673 \frac{\epsilon}{y^2} = 6.12 \times 10^{-6} \left(\frac{d_g}{10^{-6}} \right) \frac{1}{y}.$$

Thus thermal selection becomes stronger as f_a decreases, even though ϵ decreases.

8 EP-saturating parameter ridge

The optimization uses the conservative working cap

$$d_g = 10^{-6},$$

motivated by a leading pure-QCD recast of MICROSCOPE. It is not a substitute for a complete likelihood including all dilaton charges and finite-range effects.

The scan fixes

$$m_a = 7.5 \times 10^{-29} \text{ eV}$$

as a representative late-roll mass and imposes

$$T_R = 100M < f_a, \quad M > 2.6 \text{ TeV}, \quad M < 10^{12} \text{ GeV}, \quad \eta_{\Psi, \max} > 1.$$

For $N = 6$, the mass relation reduces numerically to

$$M \simeq \frac{1.002 \times 10^{-2} \text{ GeV}}{y} \left(\frac{m_a}{7.5 \times 10^{-29} \text{ eV}} \right)^{1/2} \left(\frac{10^{-6}}{d_g} \right)^{3/2}.$$

The viable scan interval is approximately

$$6.4 \times 10^{-10} \lesssim y \lesssim 3.9 \times 10^{-6},$$

or

$$1.6 \times 10^9 \text{ GeV} \lesssim f_a \lesssim 9.4 \times 10^{12} \text{ GeV},$$

with

$$2.6 \times 10^3 \text{ GeV} \lesssim M \lesssim 1.6 \times 10^7 \text{ GeV}.$$

The $N = 8$ scan survives only at much higher threshold masses,

$$4.2 \times 10^{-7} \lesssim y \lesssim 7.5 \times 10^{-6},$$

$$1.1 \times 10^8 \text{ GeV} \lesssim M \lesssim 8.2 \times 10^9 \text{ GeV}.$$

No $N = 10$ point survives the same $M < 10^{12}$ GeV, reheating, collider, and focusing cuts.

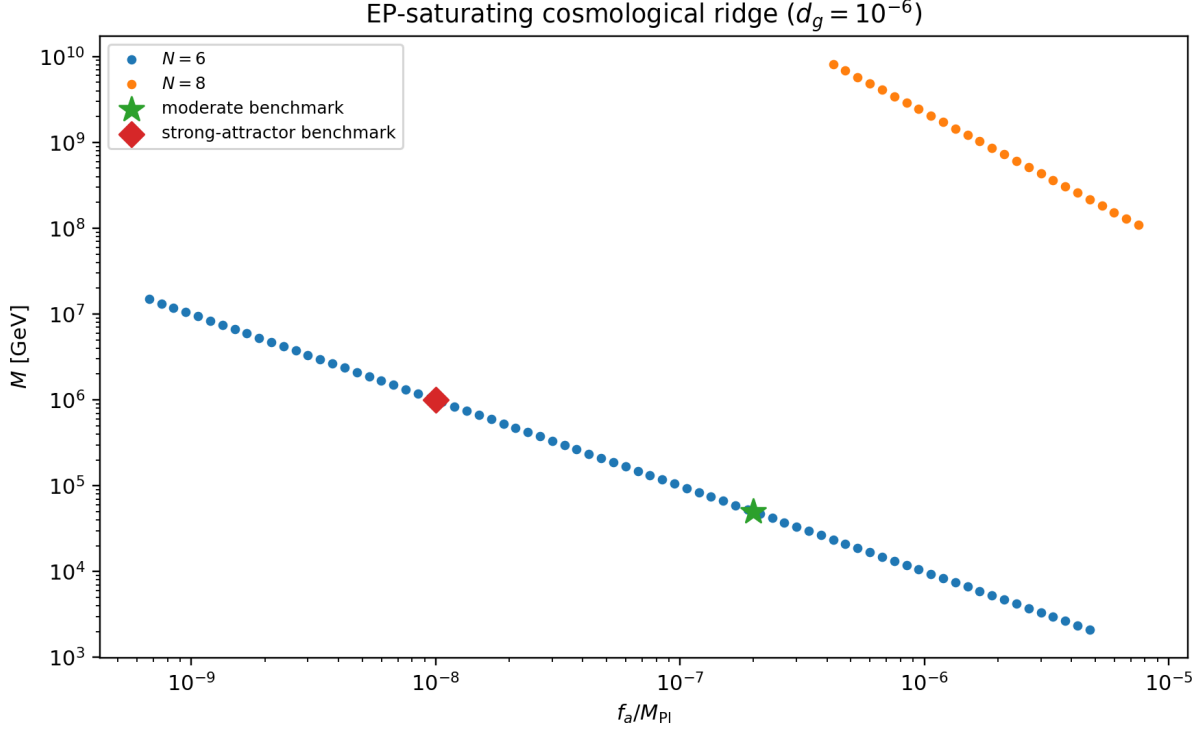


Figure 3: The largest shear is a ridge rather than a unique point. $N=6$ is parametrically and cosmologically cheaper than $N=8$.

9 Force hierarchy and chronology

The sequence is:

$$\text{reheating} \rightarrow \text{heavy-threshold focusing} \rightarrow \text{QCD refocusing} \rightarrow \text{baryon locking} \rightarrow \text{late tachyonic release} \rightarrow x_v = \pm\pi/6.$$

The QCD crossover is dynamically important even though the final linear coupling is generated by the zero-temperature threshold. It acts as a second environmental lens for the same ratio mode.

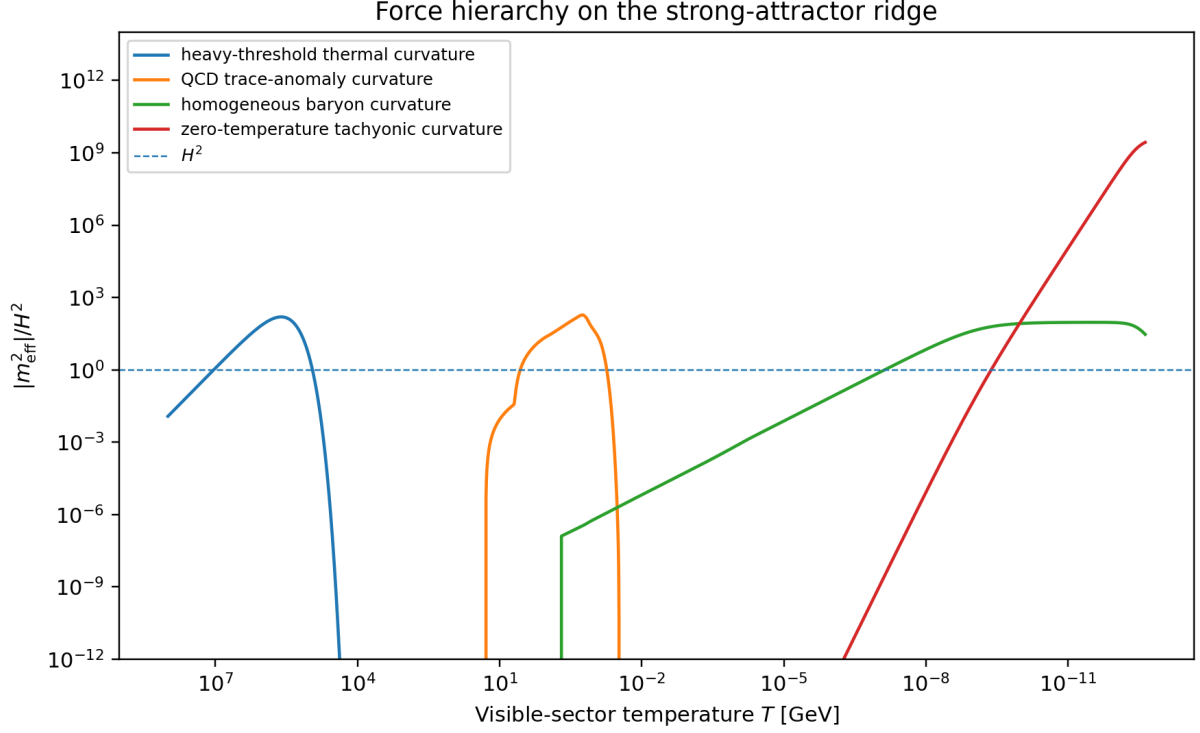


Figure 4: The heavy threshold first focuses the phase, the QCD trace anomaly refocuses it near 180 MeV, baryons hold it close to the even- N maximum, and the zero-temperature tachyon finally releases it.

10 Chronometric shear

For two clocks,

$$\mathcal{S}_{AB} = d \ln \frac{\nu_A}{\nu_B}.$$

In the pure-QCD benchmark,

$$\mathcal{S}_{AB} \simeq K_{AB} d_g d\varphi,$$

where K_{AB} is the difference in QCD/nuclear sensitivity and $\varphi = a/M_{\text{Pl}}$.

A source with scalar charge approximately d_g produces

$$\frac{a_{\text{source}}}{M_{\text{Pl}}} \simeq -2d_g \Phi_N.$$

For the annual peak-to-peak change of the solar potential,

$$\Delta\Phi_{\odot} = 3.29 \times 10^{-10},$$

and $m_a \ll \text{AU}^{-1}$,

$$\left| \Delta \ln \frac{\nu_A}{\nu_B} \right|_{\text{yr}} = 2|K_{AB}|d_g^2 \Delta\Phi_{\odot}.$$

At $d_g = 10^{-6}$,

$$\left| \Delta \ln \frac{\nu_A}{\nu_B} \right|_{\text{yr}} = 6.58 \times 10^{-22} |K_{AB}|.$$

Recent thorium-229 work uses provisional strong-sector sensitivity factors $|K_g|, |K_{\hat{m}}| = 10^4$ and 10^5 , while emphasizing that reliable ab-initio coefficients are not yet available. Taking 10^4 only as an illustrative benchmark gives

$$6.58 \times 10^{-18}.$$

This is a source-induced annual signal, not an oscillating-dark-matter signal: the scalar abundance on the optimized ridge is negligible.

The clock/EP consistency relation derived in v0.5 remains

$$\frac{\beta_{AB}}{\eta_{CD}} = -\frac{\Delta K_{AB}}{\Delta Q_{\hat{m}}'^{CD}},$$

under the one-scalar, unscreened, QCD-dominated assumptions. Cosmological vacuum selection fixes the sign of d_g , but the ratio above is independent of its magnitude.

11 Abundance and cosmological consistency

For a constant mass that begins oscillating in radiation domination,

$$\Omega_a h^2 \propto f_a^2 m_a^{1/2} \theta_i^2.$$

The old Planck-scale benchmark gives $\Omega_a h^2 \simeq 164$ for $\theta_i = \pi/6$. The moderate and strong ridge points give

$$9.2 \times 10^{-16}, \quad 2.3 \times 10^{-18},$$

respectively. The ratio mode is therefore not the dark matter in the optimized chronometric model.

The strong benchmark also satisfies

$$\frac{T_R}{f_a} = 4.11 \times 10^{-3},$$

which is compatible with avoiding radial-symmetry restoration for order-one thermal couplings, although the actual restoration temperature depends on the UV completion.

Outstanding cosmological requirements are:

- an explicit reheaton producing $\xi_0 = 1$, $\xi_5 = 0.25$, and negligible temperatures in the other replicas;
- prompt decays or acceptable relic abundances for the heavy vectorlike thresholds;
- a perturbation/isocurvature calculation rather than only homogeneous evolution;
- proof that the chosen reheating hierarchy is stable under portals between replicas;
- a permanent domain-wall cure if pre-inflation symmetry breaking is not imposed.

12 Novelty boundary

The following ingredients are established:

- nonlinearly realized Z_N protection of a light pseudo-Goldstone potential;
- replicated-sector finite-temperature scalar cosmology;
- lattice QCD thermodynamics;
- dilaton equivalence-principle charges;
- clock searches for gluon and quark-mass couplings.

The candidate new package is narrower:

protected real vectorlike-QCD threshold + $\frac{2}{27}$ transmission + entropy-asymmetric reheating phasor
 +lattice trace-anomaly evolution + baryon locking + chronometric shear/EP ridge.

The most important new negative result is also useful: moderate thermal focusing does not guarantee unique vacuum selection. The system can retain the inflationary homogeneous phase through threshold decoupling and later environmental kicks. A strong-attractor island exists, but it must be demonstrated rather than inferred from $m_T/H > 1$ alone.

13 Current verdict

Target	Verdict
Heavy-threshold thermal evolution	Pass
Independent replica entropy histories	Pass within decoupled-sector approximation
QCD crossover	Pass at leading lock defect
Matter domination	Pass
Local Earth/Sun screening	Pass from v0.7: unscreened
Unique branch at moderate focusing	Fail
Unique branch at strong benchmark	Pass for 12 tested homogeneous phases
Dark-radiation bound	Pass for one adjacent copy at $\xi = 0.25$
Misalignment abundance	Pass; negligible
Maximum nonzero chronometric shear	EP-limited at working $d_g \sim 10^{-6}$
Permanent domain-wall solution	Conditional/open
Full reheating UV completion	Open
Absolute priority claim	Not yet established

The preferred cosmological statement is now:

N=6 admits a low- f_a attractor ridge on which the Universe selects a nonzero QCD chronometric-shear vacuum.

It is not the origin of causal order. It is a concrete mechanism by which cosmological history selects which small, non-universal spectral mismatch survives after a universal clock scale has formed.

14 References

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